UE26MA141A Engineering Mathematics – I · Unit 1 · Partial Differentiation

Homogeneous Functions and Euler's Theorem

Complete study pack for class 005: homogeneous functions and their degree, Euler's theorem, the second- and higher-order corollary, the extension for f(u), and applications

Compiled from the PESU class 005 slide deck and AV-summary transcript, the Unit 1 class-work sheet and past ESA papers. Generated 14 October 2026.

· How to use this pack

Topics 1–5 follow the lecture. Class 006 (“Problems on Euler's theorem”) is a separate session and is not included; the worked problems here come from the class-work sheet and past ESA papers. Every topic has the same parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S005·6	Deck 005, slide 6 (rendered .pptx)
T005 00:08:57	AV-summary transcript 005, segment start
CW U1 P20	Unit 1 class-work problem 20
ESA Jan-24 s3 1c	Past paper: Jan 2024, set 3, question 1.c

Contents

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5. 5 · Applications of homogeneous functions
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1 Homogeneous functions

A function is homogeneous of degree n if it can be written as x^n times a function of the ratios (y/x, or y/x and z/x). Equivalently, scaling every variable by t scales the function by t^n. Polynomials are homogeneous when every term has the same degree. The degree can be positive, negative, fractional or zero.

Key terms and definitions

Homogeneous function (two variables). A function $u = f(x, y)$ is said to be a homogeneous function of degree n if it can be expressed in the form $x^n g\left(\frac{y}{x}\right)$ or $y^n g\left(\frac{x}{y}\right)$, g being an arbitrary function. S005·3 · T005 00:02:45

Homogeneous function (three variables). $u = f(x, y, z)$ is homogeneous of degree n if it can be expressed as $x^n g\left(\frac{y}{x}, \frac{z}{x}\right)$, or $y^n g\left(\frac{x}{y}, \frac{z}{y}\right)$, or $z^n g\left(\frac{x}{z}, \frac{y}{z}\right)$. S005·4 · T005 00:04:38

Scaling test. f is homogeneous of degree n exactly when $f(tx, ty) = t^n f(x, y)$ for all $t > 0$: scaling every variable by t scales the function by t^n. (added for clarity) S005·5

Degree. The power n in $x^n g(y/x)$; for a polynomial whose terms all have the same total degree, that common degree. T005 00:01:31

Formulas

E1.1 · Homogeneous of degree n ESA ★★

$$u = x^n g\left(\frac{y}{x}\right) = y^n h\left(\frac{x}{y}\right) \quad \Longleftrightarrow \quad u(tx, ty) = t^n\, u(x, y)$$

gives	The definition and the equivalent scaling test.
applies	Two variables; with three, use $x^n g(y/x, z/x)$.
note	The scaling test is added for clarity; the slides use the $x^n g(y/x)$ form.

S005·3 · S005·4 · S005·5

E1.2 · Degree of a quotient and of a product ESA ★★

$$\deg \frac{P}{Q} = \deg P - \deg Q, \qquad \deg(PQ) = \deg P + \deg Q$$

gives	Degree of combinations of homogeneous functions.
applies	P, Q homogeneous.
note	$\frac{\sqrt{x}+\sqrt{y}}{x+y}$: $\frac{1}{2} - 1 = -\frac{1}{2}$. A ratio such as $\sin^{-1}(y/x)$ has degree 0.

(added for clarity) T005 00:07:50

Graphs and diagrams

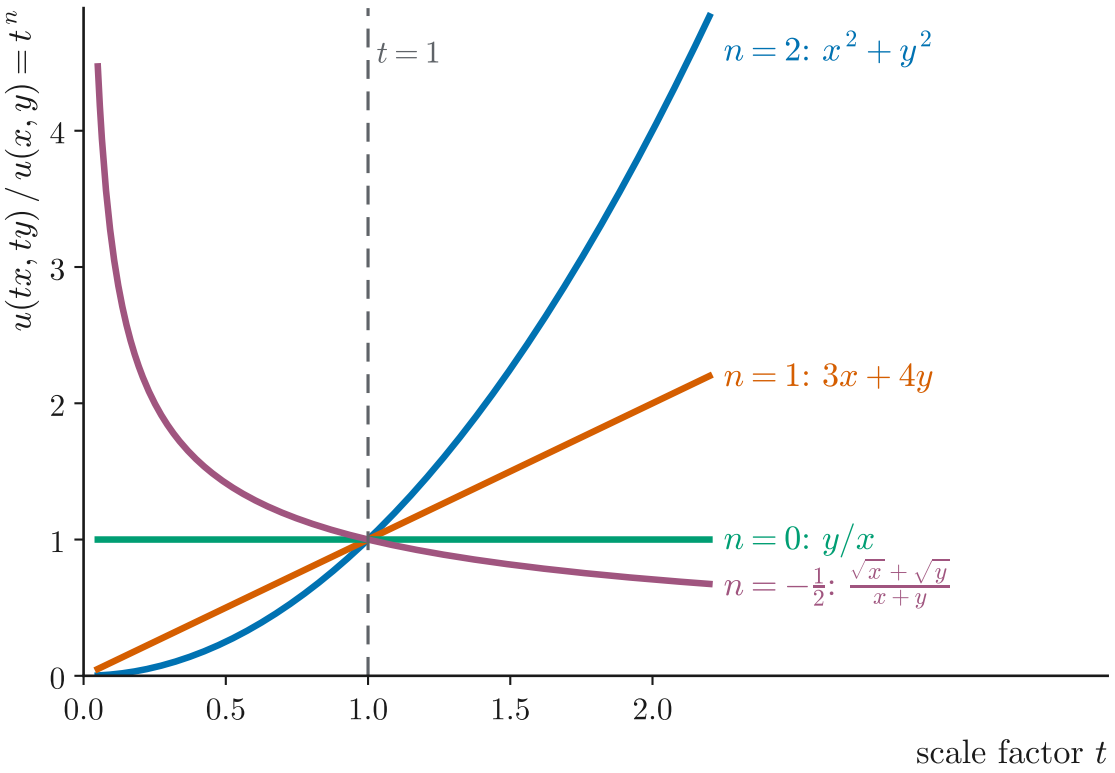


Figure G1.1. Scaling along a ray: $u(tx,ty)=t^n u(x,y)$. ESA ★★

ADDED FOR CLARITY S005·3 · S005·5

How to read it. Start at a point and move out along the ray through the origin by the factor t. A homogeneous function of degree n changes like t^n: degree 2 grows quadratically, degree 1 linearly, degree 0 stays constant along the ray (but can vary between rays), and degree $-\frac{1}{2}$ decays. Each curve is $u(t)/u(1)$.

Flowcharts

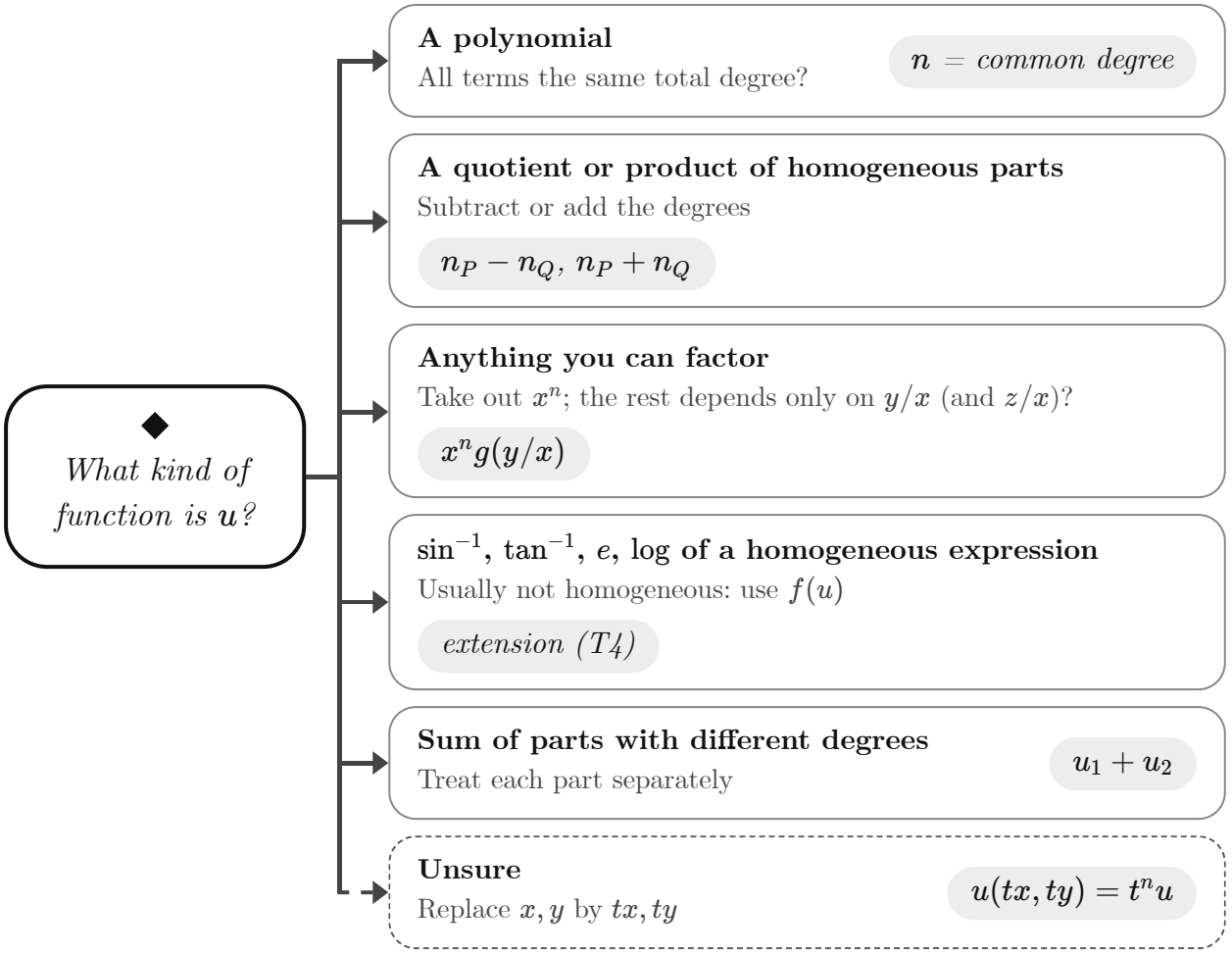


Figure F1.1. Is it homogeneous, and of what degree? ESA ★★

S005·3 · S005·4 · T005 00:02:36 · T005 00:05:42

Important points and why

1. For a polynomial, check that every term has the same total degree; that is the degree ($x^3 + 3x^2y + 3xy^2 + y^3$: degree 3). ESA ★★ T005 00:01:15 · T005 00:01:31
2. To test any function: take out x^n (or y^n) and check that what remains depends only on y/x (and z/x). ESA ★★ S005·3 · T005 00:05:42
3. Either form, $x^n g(y/x)$ or $y^n h(x/y)$, may be used — whichever is convenient — and both give the same degree. ESA ★★ T005 00:04:01
4. Inverse trigonometric or exponential functions of a homogeneous expression are usually not homogeneous; first find an f with $f(u)$ homogeneous (topic 4). ESA ★★★ T005 00:01:47 · T005 00:02:04
5. Homogeneous functions appeared earlier in homogeneous differential equations, solved by $y = vx$. ESA ★ (transcript only) T005 00:00:22

Exam trap. The lecture says the converse holds: "a function of degree 0 should be a constant function". It does not: y/x, $\sin^{-1}(y/x)$ and $\frac{x-y}{x+y}$ all have degree 0 and are not constant.

ADDED FOR CLARITY T005 00:07:11

Exam trap. "Homogeneous" does not mean "smooth everywhere" (as said at T005 00:11:06); it is a scaling property. $(\sqrt{x}+\sqrt{y})/(x+y)$ is homogeneous but not defined at the origin.

ADDED FOR CLARITY T005 00:11:06

Worked examples

W1.1 · Degree of a linear function — ESA ★

GIVEN $u = 3x + 4y$.

STEP 1 $u = x\left(3 + 4\frac{y}{x}\right) = x^1 g\left(\frac{y}{x}\right)$.

ANSWER Homogeneous of degree 1

CHECK $u(tx, ty) = t\,u$.

S005·3

W1.2 · Three variables — ESA ★★

GIVEN $u = x^3 + y^3 + z^3 + 3xyz$.

STEP 1 $u = x^3\left(1 + \left(\frac{y}{x}\right)^3 + \left(\frac{z}{x}\right)^3 + 3\frac{y}{x}\frac{z}{x}\right) = x^3 g\left(\frac{y}{x}, \frac{z}{x}\right)$.

ANSWER Homogeneous of degree 3

CHECK Every term has total degree 3. (The slide prints the second term's exponent as 2; it should be 3.)

S005·4 · T005 00:04:52

W1.3 · A negative, fractional degree — ESA ★★

GIVEN $u = \dfrac{\sqrt{x}+\sqrt{y}}{x+y}$.

STEP 1 Numerator: $\sqrt{x}\left(1+\sqrt{y/x}\right)$; denominator: $x(1+y/x)$.

STEP 2 $u = x^{1/2-1}\, g\left(\frac{y}{x}\right) = x^{-1/2} g\left(\frac{y}{x}\right)$.

ANSWER Homogeneous of degree $-\frac{1}{2}$

CHECK Euler: $xu_x + yu_y = -\frac{1}{2}u$ (sympy).

TRANSCRIPT ONLY T005 00:07:50

SHORT-NOTE RECAP

$u = f(x, y)$ is homogeneous of degree n if $u = x^n g(y/x)$ or $y^n g(x/y)$; in three variables $x^n g(y/x, z/x)$. Equivalently $u(tx, ty) = t^n u$. A polynomial is homogeneous if all its terms have the same degree. The degree may be positive, negative, fractional or zero, e.g. $(\sqrt{x} + \sqrt{y})/(x + y)$ has degree $-\frac{1}{2}$.

2 Euler's theorem

For a homogeneous function of degree n, $x\,u_x + y\,u_y = n\,u$ (and $x\,u_x + y\,u_y + z\,u_z = n\,u$ in three variables). Once you know the function is homogeneous and its degree, the combination is known without differentiating. Geometrically, the changes along x and y, scaled by x and y, add up to the function scaled by its degree.

Key terms and definitions

> *"Statement 1: If $u = f(x, y)$ is a homogeneous function of degree n then $x\dfrac{\partial u}{\partial x} + y\dfrac{\partial u}{\partial y} = nu$."* S005·6

> *"Statement 2: If $u = f(x, y, z)$ is a homogeneous function of degree n then $x\dfrac{\partial u}{\partial x} + y\dfrac{\partial u}{\partial y} + z\dfrac{\partial u}{\partial z} = nu$."* S005·6

Formulas

E2.1 · Euler's theorem, two variables ESA ★★★

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = n\,u$$

gives	The scaled sum of first partials.
applies	u homogeneous of degree n.
note	Compact form: $xu_x + yu_y = nu$.

S005·6 · T005 00:08:57

E2.2 · Euler's theorem, three variables ESA ★★

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z} = n\,u$$

gives	The same result with one term per variable.
applies	$u = f(x, y, z)$ homogeneous of degree n.
note	Extends to any number of variables.

S005·6 · T005 00:14:28

Graphs and diagrams

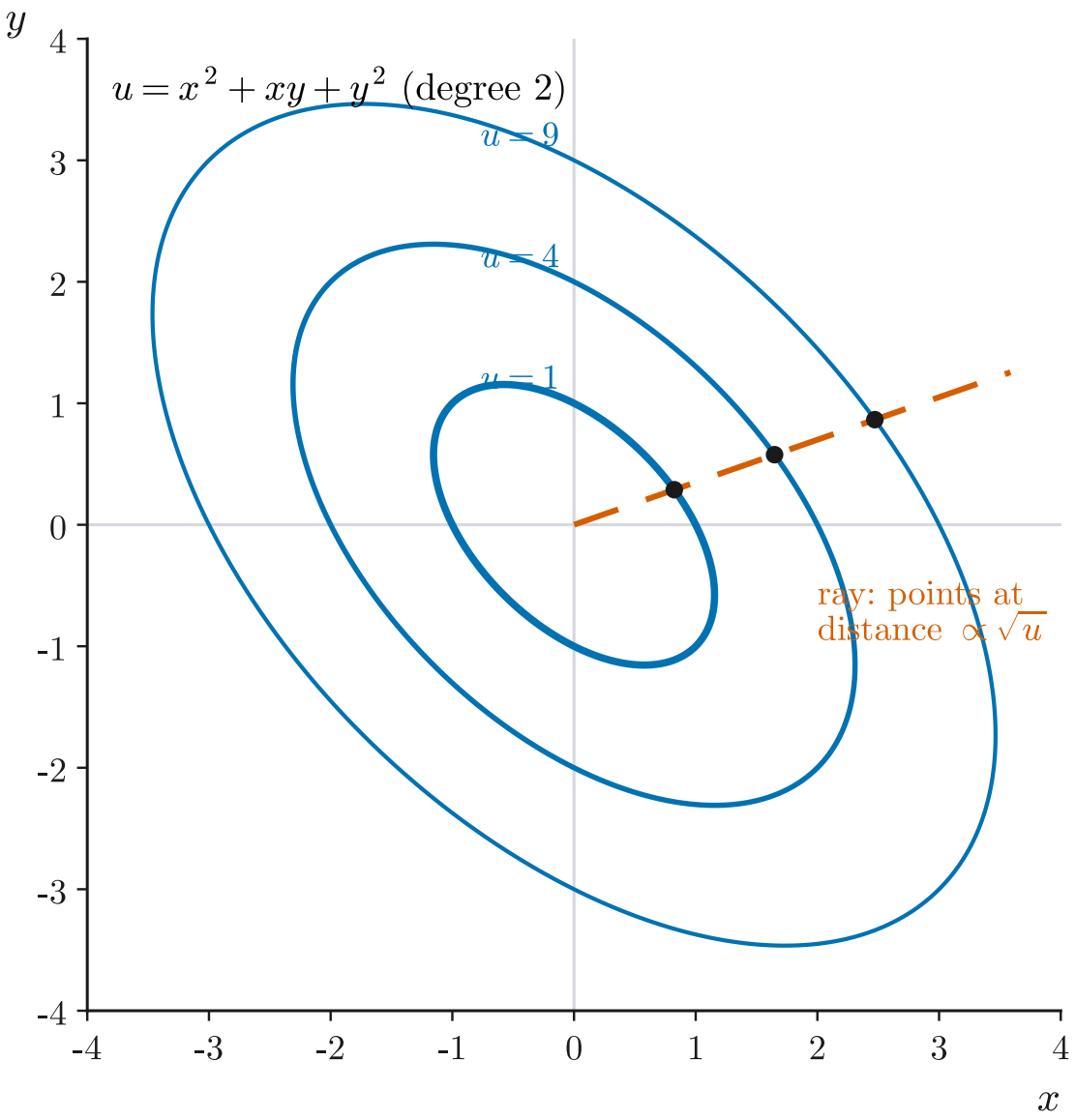


Figure G2.1. Level curves of a homogeneous function are scaled copies. ESA ★★

ADDED FOR CLARITY S005·5 · T005 00:10:18

How to read it. Level curves of $u = x^2 + xy + y^2$ (degree 2) for $u = 1, 4, 9$. Each is the first one scaled by $1, 2, 3 = \sqrt{u}$. Moving out along the dashed ray, the rate of change $xu_x + yu_y$ (the radial change scaled by distance) equals $2u$: this is Euler's theorem.

Euler's theorem and its relatives

	Applies when	Left side	Right side
Euler, first order	u homogeneous, degree n	$xu_x + yu_y$	nu
Corollary, second order	u homogeneous, degree n	$x^2u_{xx} + 2xyu_{xy} + y^2u_{yy}$	$n(n-1)u$
Both added	u homogeneous, degree n	first + second order	n^2u
Extension, first order	$f(u)$ homogeneous, degree n	$xu_x + yu_y$	$g(u) = nf(u)/f'(u)$
Extension, second order	$f(u)$ homogeneous, degree n	$x^2u_{xx} + 2xyu_{xy} + y^2u_{yy}$	$g(u)[g'(u) - 1]$
Three variables	$u(x, y, z)$ homogeneous	$xu_x + yu_y + zu_z$	nu

ADDED FOR CLARITY S005·6 · S005·7

Important points and why

1. You do not need to differentiate: check that u is homogeneous, find n, and write $xu_x + yu_y = nu$. ESA ★★★ T005 00:09:17 · T005 00:17:32
2. The lecturer calls Euler's theorem "the most important one" of the session. ESA ★★★ T005 00:08:57
3. Check: $u = x^2 + y^2$ gives $x(2x) + y(2y) = 2u$, degree 2. ESA ★★ T005 00:10:31
4. Meaning: the change along x scaled by x, plus the change along y scaled by y, equals the function scaled by its degree. ESA ★ T005 00:09:46 · T005 00:11:24

Why is $xu_x + yu_y = nu$? (A short proof.)
Differentiate $u(tx, ty) = t^n u(x, y)$ with respect to t using the chain rule: $x\,u_x(tx, ty) + y\,u_y(tx, ty) = nt^{n-1}u(x, y)$. Put $t = 1$.

ADDED FOR CLARITY S005·6

Worked examples

W2.1 · Euler's theorem checked directly ESA ★★

GIVEN $u = x^2 + y^2$.

STEP 1 $u_x = 2x$, $u_y = 2y$.

STEP 2 $xu_x + yu_y = 2x^2 + 2y^2$.

ANSWER $\boxed{xu_x + yu_y = 2u}$

CHECK Degree 2, as Euler predicts.

T005 00:10:31

W2.2 · Degree 1 with an inverse sine

ESA ★★

GIVEN $u = \sqrt{y^2 - x^2}\,\sin^{-1}(x/y)$; find $xu_x + yu_y$.

STEP 1 $\sqrt{y^2 - x^2} = y\sqrt{1 - (x/y)^2}$ has degree 1; $\sin^{-1}(x/y)$ has degree 0.

STEP 2 So $u = y^1 h(x/y)$: degree 1.

ANSWER $\boxed{xu_x + yu_y = u}$

CHECK CW answer: u.

CW U1 P21

SHORT-NOTE RECAP

Euler's theorem: if $u = f(x, y)$ is homogeneous of degree n then $xu_x + yu_y = nu$; for $f(x, y, z)$, $xu_x + yu_y + zu_z = nu$. So only homogeneity and the degree are needed, not the derivatives. Proof: differentiate $u(tx, ty) = t^n u$ with respect to t and put $t = 1$.

3 Second and higher order: the corollary

Applying Euler's idea twice gives $x^2u_{xx} + 2xy\,u_{xy} + y^2u_{yy} = n(n-1)\,u$, and three times gives $n(n-1)(n-2)\,u$. When u is a sum of homogeneous parts of different degrees, apply the results to each part separately. This is the commonest ESA question on the topic.

Key terms and definitions

> *"Corollary 1: If $u = f(x,y)$ is a homogeneous function of degree n then $x^2\dfrac{\partial^2 u}{\partial x^2} + 2xy\dfrac{\partial^2 u}{\partial x \partial y} + y^2\dfrac{\partial^2 u}{\partial y^2} = n(n-1)u$."* S005·6

Formulas

E3.1 · Corollary: second order ESA ★★★

$$x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x\,\partial y} + y^2\frac{\partial^2 u}{\partial y^2} = n(n-1)\,u$$

gives	The scaled sum of second partials.
applies	u homogeneous of degree n with continuous second partials.
note	Coefficients $1, 2, 1$ as in $(x+y)^2$.

S005·6 · T005 00:12:21

E3.2 · Third order ESA ★

$$x^3u_{xxx} + 3x^2y\,u_{xxy} + 3xy^2u_{xyy} + y^3u_{yyy} = n(n-1)(n-2)\,u$$

gives	The pattern continues with binomial coefficients.
applies	Third partials continuous.
note	The lecture states the right-hand side; the coefficients 1, 3, 3, 1 are added for clarity.

transcript only · added for clarity · T005 00:12:43 · T005 00:14:12

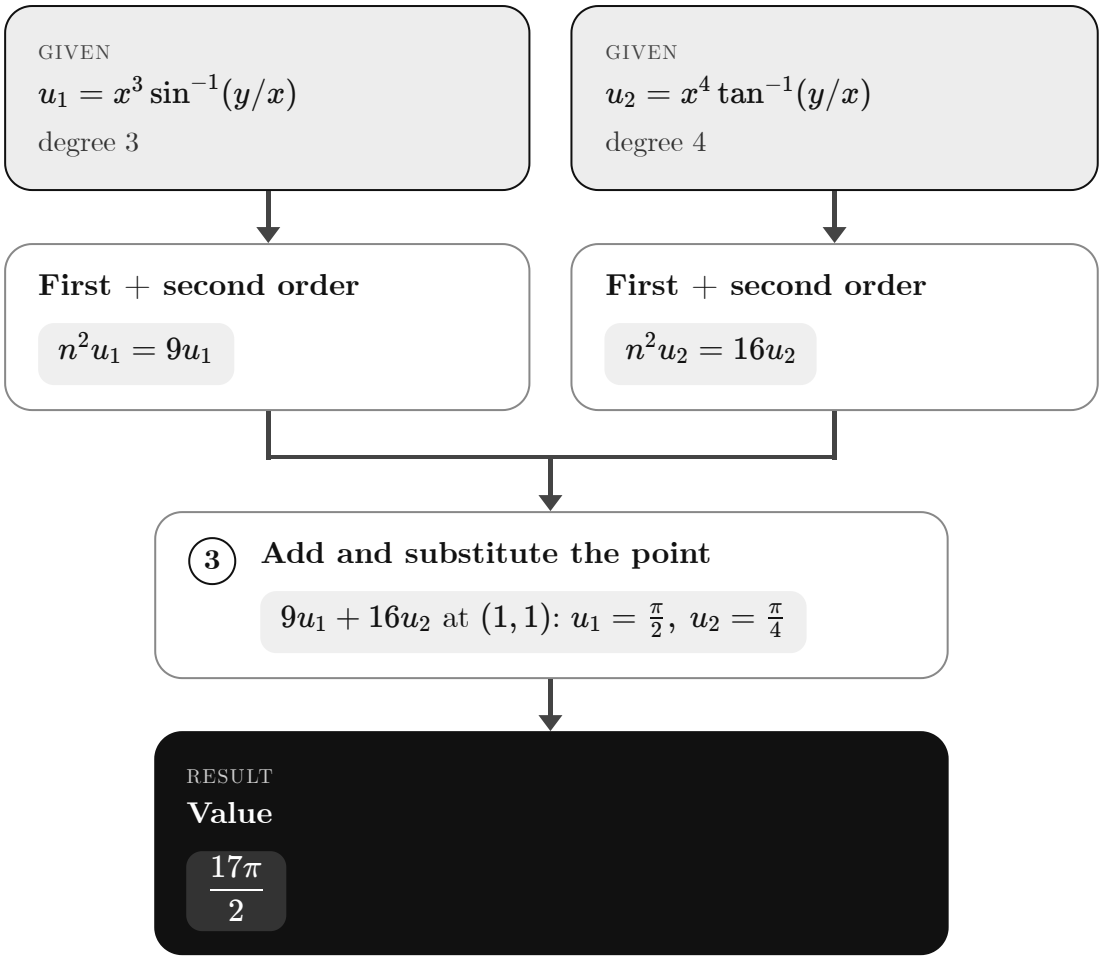
E3.3 · First plus second order ESA ★★★

$$x^2u_{xx} + 2xy\,u_{xy} + y^2u_{yy} + x\,u_x + y\,u_y = n(n-1)u + nu = n^2u$$

gives	The combination asked in ESA Jan 2024 sets 3 and 4.
applies	u homogeneous of degree n; for $u = u_1 + u_2$ apply to each part.
note	Degrees 3 and -3 both give 9: so $9u$ for $x^3\sin^{-1}\frac{y}{x} + y^{-3}\tan^{-1}\frac{x}{y}$.

added for clarity · ESA Jan-24 s3 1c · ESA Jan-24 s4 1d · CW U1 P23

Flowcharts



Remark. Only first order: n; only second order: $n(n-1)$; both: n^2. Read which combination the question asks for.

Figure F3.1. Evaluating Euler combinations for a sum of homogeneous parts. ESA ★★★

ESA Jan-24 s3 1c · ESA Jan-24 s4 1d

Important points and why

1. The mixed partials u_{xy} and u_{yx} are equal here (continuous derivatives), which is why the middle term is $2xy\,u_{xy}$. ESA ★★ T005 00:12:04
2. For $u = u_1 + u_2$ with different degrees n_1, n_2, apply the results to each part: $x^2 u_{xx} + \cdots = n_1(n_1 - 1)u_1 + n_2(n_2 - 1)u_2$. ESA ★★★ ADDED FOR CLARITY ESA Jan-24 s3 1c · CW U1 P22
3. Extend to three or more variables as an exercise; the lecture leaves it open and points to a research paper on higher-order extensions. ESA ★ S005·6 · T005 00:14:59

Why is the second-order right side $n(n-1)u$?
u_x and u_y are homogeneous of degree $n-1$, so Euler's theorem applied to them gives $xu_{xx} + yu_{xy} = (n-1)u_x$ and $xu_{yx} + yu_{yy} = (n-1)u_y$. Multiply by x and y, add, and use $xu_x + yu_y = nu$.

ADDED FOR CLARITY S005·6

Worked examples

W3.1 · Prove 5u and 20u — ESA ★★★

GIVEN $u = x^3y^2 \sin^{-1}(y/x)$. Prove (i) $xu_x + yu_y = 5u$, (ii) $x^2u_{xx} + 2xy\,u_{xy} + y^2u_{yy} = 20u$.

STEP 1 $x^3y^2 = x^5\left(\frac{y}{x}\right)^2$ and $\sin^{-1}(y/x)$ depends only on y/x, so $u = x^5 g(y/x)$: degree 5.

STEP 2 (i) Euler: $5u$.

STEP 3 (ii) Corollary: $n(n-1)u = 5 \cdot 4\,u$.

ANSWER (i) $5u$, (ii) $20u$

CHECK Direct differentiation in sympy gives the same.

ESA Jul-23 1a

W3.2 · Fractional degree, both orders — ESA ★★

GIVEN $u = \dfrac{y^{1/3} - x^{1/3}}{x + y}$.

STEP 1 Degree $\frac{1}{3} - 1 = -\frac{2}{3}$.

STEP 2 (i) $xu_x + yu_y = -\frac{2}{3}u$.

STEP 3 (ii) $n(n-1) = \left(-\frac{2}{3}\right)\left(-\frac{5}{3}\right) = \frac{10}{9}$.

ANSWER (i) $-\frac{2}{3}u$, (ii) $\frac{10}{9}u$

CHECK CW answers agree.

CW U1 P19

W3.3 · Sum of two degree-3 parts — ESA ★★

GIVEN $u = x^3 \sin^{-1}(y/x) + y^3 \tan^{-1}(x/y)$; find $x^2u_{xx} + 2xy\,u_{xy} + y^2u_{yy}$.

STEP 1 Both parts are homogeneous of degree 3, so u is.

STEP 2 $n(n-1) = 3 \cdot 2$.

ANSWER $6u$

CHECK CW answer: $6u$.

CW U1 P22

W3.4 · Degrees 3 and −3 together ESA ★★★

GIVEN $u = x^3 \sin^{-1}(y/x) + y^{-3}\tan^{-1}(x/y)$; evaluate $x^2u_{xx} + 2xy\,u_{xy} + y^2u_{yy} + xu_x + yu_y$.

STEP 1 $u_1 = x^3\sin^{-1}(y/x)$: degree 3; $u_2 = y^{-3}\tan^{-1}(x/y)$: degree -3.

STEP 2 For each part the sum is $n(n-1)+n = n^2$: $9u_1$ and $9u_2$.

ANSWER $\boxed{9u}$

CHECK CW answer: $9u$; sympy agrees.

ESA Jan-24 s4 1d · CW U1 P23

W3.5 · Degrees 3 and 4, value at (1, 1) ESA ★★★

GIVEN $u = x^3\sin^{-1}(y/x) + x^4\tan^{-1}(y/x)$; find $x^2u_{xx} + 2xy\,u_{xy} + y^2u_{yy} + xu_x + yu_y$ at $x = 1, y = 1$.

STEP 1 $u_1 = x^3\sin^{-1}(y/x)$ (degree 3) gives $9u_1$; $u_2 = x^4\tan^{-1}(y/x)$ (degree 4) gives $16u_2$.

STEP 2 At $(1,1)$: $u_1 = \sin^{-1}1 = \frac{\pi}{2}$, $u_2 = \tan^{-1}1 = \frac{\pi}{4}$.

STEP 3 $9\cdot\frac{\pi}{2} + 16\cdot\frac{\pi}{4}$.

ANSWER $\boxed{\dfrac{17\pi}{2}}$

CHECK sympy: the expression equals $9u_1 + 16u_2$ identically.

ESA Jan-24 s3 1c

Past-paper questions

ESA Jul 2023 (6 marks) UE22MA141A Q1.a · solved in W3.1

If $u(x,y) = x^3y^2\sin^{-1}\left(\frac{y}{x}\right)$, prove that (i) $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = 5u$, (ii) $x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x\partial y} + y^2\frac{\partial^2 u}{\partial y^2} = 20u$.

ESA Jan 2024 (6 marks) UE23MA141A set 3, Q1.c · solved in W3.5

If $u = x^3\sin^{-1}\left(\frac{y}{x}\right) + x^4\tan^{-1}\left(\frac{y}{x}\right)$, find $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} + xu_x + yu_y$ at $x = 1$, $y = 1$ by using Euler's theorem and its extension.

ESA Jan 2024 (6 marks) UE23MA141A set 4, Q1.d · solved in W3.4

If $u = x^3\sin^{-1}\left(\frac{y}{x}\right) + y^{-3}\tan^{-1}\left(\frac{x}{y}\right)$, evaluate $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} + xu_x + yu_y$ by using Euler's theorem and its extension.

SHORT-NOTE RECAP

Corollary: $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} = n(n-1)u$; adding the first-order result gives n^2u. For a sum of homogeneous parts of different degrees apply this to each part, e.g. degrees 3 and 4 give $9u_1 + 16u_2$. Third order: $n(n-1)(n-2)u$.

4 Extension: when f(u) is homogeneous

Functions such as $\tan^{-1}\frac{x^4+y^4}{xy}$ are not homogeneous, but $\tan u$ is. If $f(u)$ is homogeneous of degree n, then $x\,u_x + y\,u_y = g(u) = n\,\dfrac{f(u)}{f'(u)}$ and $x^2u_{xx} + 2xy\,u_{xy} + y^2u_{yy} = g(u)\,[g'(u) - 1]$. This extension is on the last content slide and is asked in two ESA papers.

Key terms and definitions

> *"If $f(u)$ be a homogeneous function of degree n in x and y, then $x\dfrac{\partial u}{\partial x} + y\dfrac{\partial u}{\partial y} = n\dfrac{f(u)}{f'(u)}$ and $x^2\dfrac{\partial^2 u}{\partial x^2} + 2xy\dfrac{\partial^2 u}{\partial x\partial y} + y^2\dfrac{\partial^2 u}{\partial y^2} = g(u)[g'(u) - 1]$ where $g(u) = n\dfrac{f(u)}{f'(u)}$."* S005·7

Formulas

E4.1 · Extension, first order ESA ★★★

$$f(u) \text{ homogeneous of degree } n \implies x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = n\frac{f(u)}{f'(u)} = g(u)$$

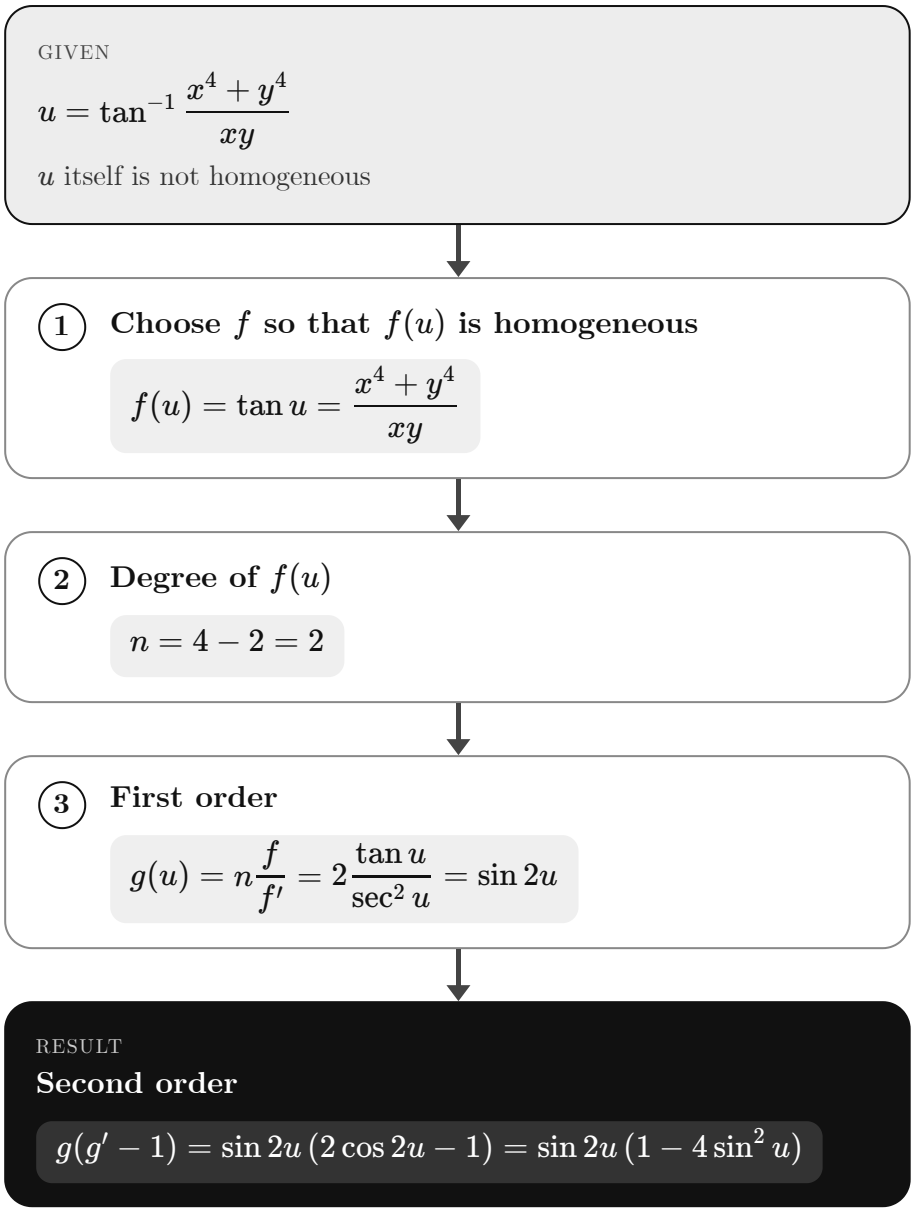
GIVES Euler's combination for a function that is not itself homogeneous.
APPLIES Some $f(u)$ is homogeneous (e.g. $\tan u$, $\sin u$, e^u).
NOTE $\tan u$: $g = n\tan u\cos^2 u = \frac{n}{2}\sin 2u$.
S005·7

E4.2 · Extension, second order ESA ★★★

$$x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x\,\partial y} + y^2\frac{\partial^2 u}{\partial y^2} = g(u)\left[g'(u) - 1\right]$$

GIVES The second-order combination for the extension.
APPLIES $g(u) = n\,f(u)/f'(u)$ from E4.1.
NOTE Differentiate g with respect to u.
S005·7

Flowcharts



Remark. Common choices: $\sin^{-1}(\dots) \to f = \sin$ ($g = n\tan u$); $\tan^{-1} \to f = \tan$ ($g = \frac{n}{2}\sin 2u$); $\log(\dots) \to f = e^u$ ($g = n$); $e^{(\cdots)} \to f = \log$ ($g = nu\log u$).

Figure F4.1. Using the extension of Euler's theorem. ESA ★★★

S005·7 · ESA May-24 1a

Important points and why

1. Spot the extension case: $u = \sin^{-1}(\dots)$, $\tan^{-1}(\dots)$, $\log(\dots)$ or $e^{(\cdots)}$ of a homogeneous expression; then f is $\sin$, $\tan$, $\exp$ or $\log$ respectively. ESA ★★★ ADDED FOR CLARITY S005·7 · ESA May-24 1a
2. The degree n belongs to $f(u)$, not to u. ESA ★★★ S005·7

Where does $n\,f(u)/f'(u)$ come from?
Apply Euler to $F = f(u)$: $xF_x + yF_y = nF$. By the chain rule $F_x = f'(u)u_x$, $F_y = f'(u)u_y$, so $f'(u)(xu_x + yu_y) = nf(u)$.

ADDED FOR CLARITY S005·7

Exam trap. For $u = \tan^{-1}\frac{x^4+y^4}{xy}$ the degree is that of $\tan u = \frac{x^4+y^4}{xy}$, i.e. $4-2=2$, not of u.

ADDED FOR CLARITY ESA May-24 1a

Worked examples

W4.1 · Extension with tan u

ESA ★★★

GIVEN $u = \tan^{-1}\dfrac{x^4+y^4}{xy}$. Prove (i) $xu_x + yu_y = \sin 2u$, (ii) $x^2u_{xx} + 2xy\,u_{xy} + y^2u_{yy} = \sin 2u\,(1 - 4\sin^2 u)$.

STEP 1 $f(u) = \tan u = \dfrac{x^4+y^4}{xy}$ is homogeneous of degree $4-2=2$.

STEP 2 (i) $g(u) = 2\dfrac{\tan u}{\sec^2 u} = 2\sin u\cos u = \sin 2u$.

STEP 3 (ii) $g'(u) = 2\cos 2u$, so $g(g'-1) = \sin 2u\,(2\cos 2u - 1) = \sin 2u\,(1 - 4\sin^2 u)$, using $\cos 2u = 1 - 2\sin^2 u$.

ANSWER (i) $\sin 2u$, (ii) $\sin 2u\,(1 - 4\sin^2 u)$

CHECK sympy: residuals 0 and $\approx 10^{-16}$.

ESA May-24 1a · ESA Jun-24 1b · CW U1 P20

Past-paper questions

ESA May 2024 · 6 MARKS · UE22MA141A Q1.a · solved in W4.1

If $u = \tan^{-1}\left(\frac{x^4+y^4}{xy}\right)$, prove that (i) $xu_x + yu_y = \sin 2u$, (ii) $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} = \sin 2u(1 - 4\sin^2 u)$.

ESA June 2024 · 6 MARKS · UE23MA141A Q1.b · solved in W4.1

Same question as May 2024 Q1.a: $u = \tan^{-1}\left(\frac{x^4+y^4}{xy}\right)$, prove (i) and (ii).

SHORT-NOTE RECAP

If $f(u)$ (not u) is homogeneous of degree n: $xu_x + yu_y = g(u) = nf(u)/f'(u)$ and $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} = g(u)[g'(u) - 1]$. For $u = \tan^{-1}\frac{x^4+y^4}{xy}$: $f = \tan$, $n = 2$, $g = \sin 2u$, second order $\sin 2u(1 - 4\sin^2 u)$.

5 Applications of homogeneous functions

Euler's theorem relates a homogeneous function to its partial derivatives and describes how the function scales. It is used in economics (production functions, marginal productivity, elasticity of demand) and in thermodynamics, where extensive variables are homogeneous of degree 1 and intensive variables of degree 0. A function of ratios only has degree 0, so its Euler combination is 0.

Key terms and definitions

Extensive / intensive variable. Extensive variables (mass, volume, energy) double when the system is doubled: homogeneous of degree 1. Intensive variables (pressure, temperature, density) do not change: degree 0. ADDED FOR CLARITY S005·5 · T005 00:16:26

> *"A constant function is homogeneous of degree 0. Linear functions are homogeneous of degree 1. The degree of the homogeneous function can be positive, negative or zero."* S005·5

Important points and why

1. A constant function is homogeneous of degree 0; linear functions are homogeneous of degree 1. ESA ★ S005·5 · T005 00:06:52
2. Euler's theorem also describes the scaling of the function: $n\,u$. ESA ★ S005·5 · T005 00:15:57
3. Applications: production functions, marginal productivities and elasticity of demand in economics; extensive (degree 1) and intensive (degree 0) variables in thermodynamics. ESA ★ S005·5 · T005 00:16:09
4. Any function of ratios only, such as $f(x/y,\, y/z,\, z/x)$, has degree 0, so $xu_x + yu_y + zu_z = 0$ (ESA Jan 2024 set 1). ESA ★★★ ADDED FOR CLARITY ESA Jan-24 s1 1d

Exam trap. The lecture swaps the examples: mass is extensive (degree 1); pressure and temperature are intensive (degree 0). ADDED FOR CLARITY T005 00:16:46

Worked examples

W5.1 · A function of ratios only ESA ★★★

GIVEN $u = f\left(\frac{x}{y}, \frac{y}{z}, \frac{z}{x}\right)$; find $xu_x + yu_y + zu_z$.

STEP 1 Scaling x, y, z by t leaves every ratio unchanged, so $u(tx, ty, tz) = u$: degree 0.

STEP 2 Euler in three variables: $xu_x + yu_y + zu_z = 0 \cdot u$.

ANSWER $\boxed{xu_x + yu_y + zu_z = 0}$

CHECK Same answer as the chain-rule method (sympy).

ADDED FOR CLARITY ESA Jan-24 s1 1d

Past-paper questions

ESA Jan 2024 5 marks UE23MA141A set 1, Q1.d · solved in W5.1

If $u = f\left(\frac{x}{y}, \frac{y}{z}, \frac{z}{x}\right)$, find $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z}$ (degree 0; also solvable by the chain rule).

short-note recap

Constants are homogeneous of degree 0 and linear functions of degree 1. Euler's theorem gives the scaling of a function; it is used in economics (production functions, marginal productivity, elasticity) and in thermodynamics (extensive variables degree 1, intensive degree 0). A function of ratios has degree 0, so its Euler sum is 0.

A Symbol glossary

Symbol	Name	Meaning	Appears in
$u = f(x, y)$	Function	The function tested for homogeneity	E1.1, E2.1
n	Degree	Power in $x^n g(y/x)$; any real number	E1.1–E4.2
$g(\cdot)$ (topic 1)	Arbitrary function of ratios	$u = x^n g(y/x)$	E1.1
$g(u)$ (topic 4)	Extension function	$nf(u)/f'(u)$ (a different g from topic 1)	E4.1, E4.2
$f(u)$	Homogeneous function of u	e.g. $\tan u$ when $u = \tan^{-1}(\dots)$	E4.1
t	Scale factor	$u(tx, ty) = t^n u$	E1.1
u_x, u_y, u_z	First partials	Homogeneous of degree $n - 1$	E2.1, E2.2
u_{xx}, u_{xy}, u_{yy}	Second partials	Degree $n - 2$; $u_{xy} = u_{yx}$	E3.1
u_1, u_2	Homogeneous parts	Parts of different degrees in a sum	E3.3
P, Q	Homogeneous factors	Numerator and denominator of a quotient	E1.2

B Sources, coverage and gaps

Source	Path	Use
Deck 005 (.pptx rendered; .slides.md; .slides.txt)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/005_Homogeneous_functions_and_Euler_s_theorem__b933003cc540ac04.*	8 slides; content on 3–7
Transcript 005 (all 4 formats)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/005_Homogeneous_functions_and_Euler_s_theorem__a0c2c43d28f6ec69.*	19.4 min lecture
Unit 1 & 2 class-work sheet (UE24MA141A)	pesuc/First_Year_8enterprise/Files/Math 1/CW/Math_Unit1&2_CW__1jKDN6W8iw.pdf	Problems 19–23 (with answers)
ESA papers, UE22MA141A and UE23MA141A (8 papers)	pesuc/First_Year_8enterprise/Files/Math 1/Papers/ESA Papers/UE2[23]MA141A/*.pdf	6 in-scope Q1 parts
Video 005 (.mp4)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/005_*.mp4	Not opened: slide 7 is readable in the rendered image

Eight past ESA papers were checked: UE22MA141A Dec 2023, Jul 2023 and May 2024, and UE23MA141A Jan 2024 sets 1–4 and June 2024. Euler's theorem appears in six of them: Jul 2023 Q1.a, May 2024 Q1.a, June 2024 Q1.b (the same question as May 2024), Jan 2024 set 3 Q1.c, set 4 Q1.d, and set 1 Q1.d (a degree-0 function, also solvable by the chain rule). Five are 6-mark questions, so this is one of the most examined topics in Unit 1. The definition, applications and third-order result were not asked.

Known gaps, conflicts and corrections

1. Slide 005·7 (the extension of Euler's theorem) has its equations only as images; the .slides.md text shows just the title. They were read from the rendered slide.
2. Slide 005·4 writes $x^3\left(1+(y/x)^2+(z/x)^3+3\frac{y}{x}\frac{z}{x}\right)$; the second exponent should be 3.
3. The lecture (T005 00:07:11) says that a function of degree 0 must be constant. That is false (y/x has degree 0); corrected as an exam trap.
4. The lecture (T005 00:11:06) calls homogeneous functions “smooth everywhere”; homogeneity is a scaling property, not smoothness.
5. The lecture (T005 00:16:46) swaps the thermodynamics examples: mass is extensive (degree 1); pressure and temperature are intensive (degree 0).
6. The lecture's third-order list of partial derivatives (T005 00:12:43) is incomplete; the result $n(n-1)(n-2)u$ is stated with the binomial coefficients added for clarity.
7. Biographical remarks about Euler in the lecture are left out (not examinable, and partly inaccurate).
8. Class 006 (“Problems on Euler's theorem”, two transcripts) is a separate session and is not included.

9. Transcript recognition errors were corrected silently: “oiless theorem” → Euler's theorem, “Euless theorem” → Euler's theorem, “quintiple” → quintic, “y bytes” → y/x.

Coverage

Source	Citations	Slides cited (slide × citations)
S005	37	3×6, 4×4, 5×9, 6×10, 7×8
CW	7	
ESA	15	
T005	37	

Slides 005·1, 005·2 and 005·8 are title, session and closing slides. Slide 005·7's equations exist only in the slide image; the text export loses them.